SC4001 · Part B

YourName | Neural Networks and Deep Learning | October 11, 2026

Generative AI declaration

I used OpenAI Codex / ChatGPT for planning, code generation/debugging, experiment execution, analysis, figures, report drafting and submission packaging. Verification included assignment/Q&A/library-source checks, behavioral tests, CUDA checkpoint restoration, saved-prediction metric/BCE checks, independent parameter/FLOPs formulas and artifact hashes. Grad-CAM tensors were reproduced by independent PyTorch autograd. All results below come from completed saved experiments. I remain responsible for reviewing and understanding the submitted work.

Data, architecture interpretation and common protocol

Use the supplied 4,708/524/624 train/validation/test partitions and 64×64 grayscale images. Divide uint8 by 255 once, then normalize with training-pixel mean 0.5716994213 and population SD 0.1740975398. Q&A confirms 0=healthy, 1=pneumonia. Retain these original-training normalization statistics during final refits, following the brief; the FAQ only requires preprocessing refits for A1. Eight distinct images overlap train/validation; three extra copies occur within test. No exact train/test or validation/test matches were found, but patient independence cannot be checked without IDs.

Convolutions have no bias, then BN/ReLU; FC layers have bias; residual sums use ReLU. Q&A confirms SAME padding=kernel_size//2, including stride 2. It corrects the caption's 'all layers': B1/B2 final FC1 has an unrestricted logit with BCEWithLogitsLoss, no ReLU/dropout or prior sigmoid. Apply the same binary-loss interpretation to B3 (not separately specified in that answer). Sigmoid≥0.5 predicts class 1; B1 dropout 0.5 follows both hidden FC ReLUs, B2/B3 use none.

Selection: Adam, seed 42, 20-epoch cap, patience 4, minimum validation-loss improvement 0.0001; save lowest validation error, then lower loss, then earlier epoch. Candidate ties prefer smaller rate/batch. Per Q&A, initialize each final model fresh and refit on all 5,232 train+validation images for its selected 9/13/9 epochs (B1/B2/B3), with frozen hyperparameters, then evaluate test once. Fixed refits use no validation/test stopping. No augmentation, weighting or resampling; metrics use evaluation mode.

B1. AlexNet-like learning-rate selection

Figure B1 is implemented as 5×5/16 → max pool → 5×5/32 → max pool → 3×3/64 → max pool → average output 4×4 → FC64 → FC32 → FC1, with dropout 0.5 only in hidden FC layers. All five rates use batch 128 and fresh initialization; 99,569 trainable parameters.

Learning rate	Best epoch	Val. BCE loss	Val. error	Sensitivity	Specificity
1e-05	18	0.283130	9.5420%	97.1722%	71.1111%
1e-04	19	0.076439	3.0534%	97.4293%	95.5556%
1e-03	**9**	**0.064911**	**2.0992%**	**98.2005%**	**97.0370%**
1e-02	5	0.199377	5.9160%	98.7147%	80.7407%
1e-01	17	0.103817	4.5802%	96.1440%	93.3333%

Select LR=0.001, epoch 9: 11/524 validation errors (97.9008% accuracy), versus 16 for the next-best rate 0.0001. Every metric in a row belongs to that one saved checkpoint; stopping loss and error selection can prefer different epochs. The frozen learning rate supplies B2 and B3.

Why sensitivity and specificity are necessary

Validation is 74.24% positive: predicting pneumonia for everyone would yield 74.24% accuracy, 100% sensitivity and 0% specificity. Sensitivity=TP/(TP+FN) measures detected positives; specificity=TN/(TN+FP) measures correctly rejected negatives. LR 0.01 has higher sensitivity than 0.001 but far lower specificity (80.74% versus 97.04%), a trade-off hidden by one aggregate score.

B1 · learning curves and test; B2 selection table

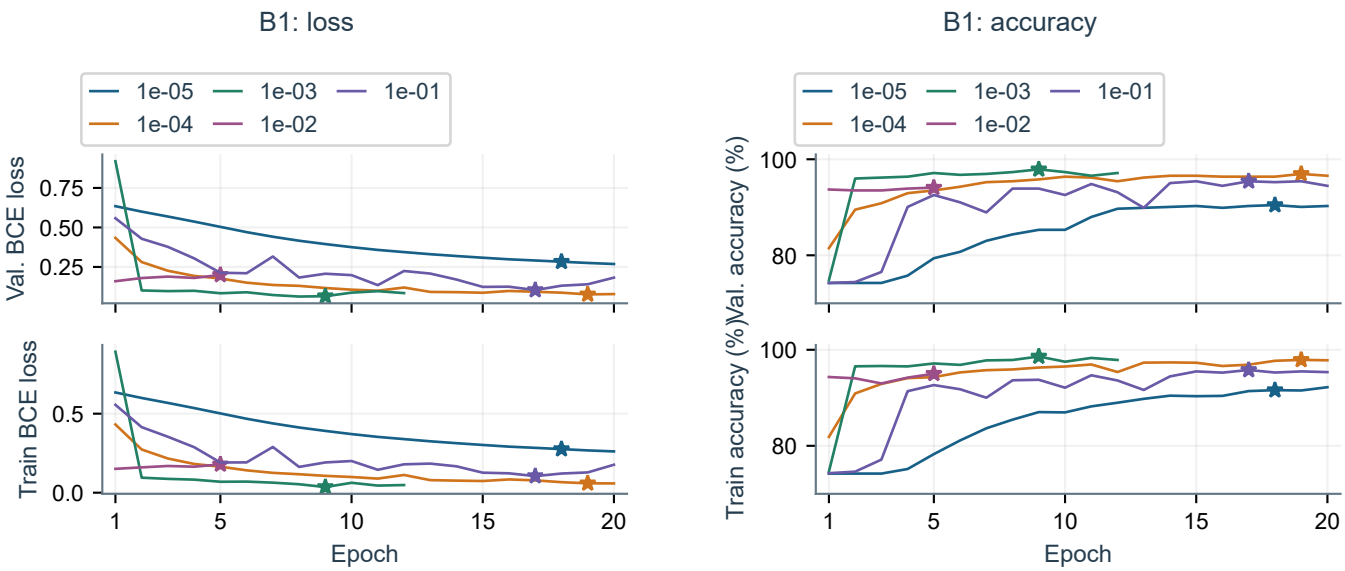


Figure B1a–b. Separate loss and accuracy figures, each with all five validation and training curves. Stars mark saved checkpoints. Accuracy axes start at 70%; curves end at actual stopping epochs.

LR 1e-5 learns slowly: loss keeps falling at the 20-epoch cap, but best accuracy is only 90.46%. 1e-4 improves more gradually to 96.95% at epoch 19. 1e-3 reaches minimum loss at epoch 8, but epoch 9 has lower error; patience stops at 12. Its selected train/validation losses are 0.035620/0.064911. 1e-2 worsens in both train/validation loss and stops at 5, indicating an unfavorable optimization trajectory rather than a simple overfitting pattern. 1e-1 shows reversals and reaches only 95.42% best accuracy. Finite-budget stopping is not proof of global convergence; BN/dropout causes were not isolated.

B1. One final test evaluation

Test accuracy	Sensitivity	Specificity	TP / TN / FP / FN
90.7051%	97.6923%	79.0598%	381 / 185 / 49 / 9

The fresh nine-epoch combined-data refit misses 9 positives and labels 49/234 negatives positive. Test accuracy is 7.20 percentage points below selection validation accuracy; test specificity is 79.0598%. Prevalence changes affect accuracy but cannot alone explain lower specificity. Supplied overlap/dataset differences and refitting may contribute; their effects are not isolated. No settings or threshold were changed after test.

B2. VGG-like batch-size selection, frozen LR=0.001

Batch size	Best epoch	Val. BCE loss	Val. error	Sensitivity	Specificity
16	**13**	**0.069390**	**2.0992%**	**98.9717%**	**94.8148%**
32	13	0.090230	2.8626%	96.9152%	97.7778%
64	6	0.097501	2.8626%	98.7147%	92.5926%
128	5	0.135308	4.7710%	98.7147%	85.1852%
256	16	0.085114	2.8626%	98.2005%	94.0741%

Batch 16 wins uniquely with 11 errors, versus 15 for 32/64/256 and 25 for 128. Batch 32 has higher specificity, but the prescribed lowest-error criterion selects batch 16. LR/batch are frozen for B3.

B2 · batch-size curves, interpretation and final test

The VGG-like model has two 3×3 Conv/BN/ReLU layers per width 8,16,32, each pair followed by max pooling, then global average pooling and FC1: 18,185 parameters, no dropout. All candidates share initial tensors and the underlying shuffled row sequence; batch grouping changes.

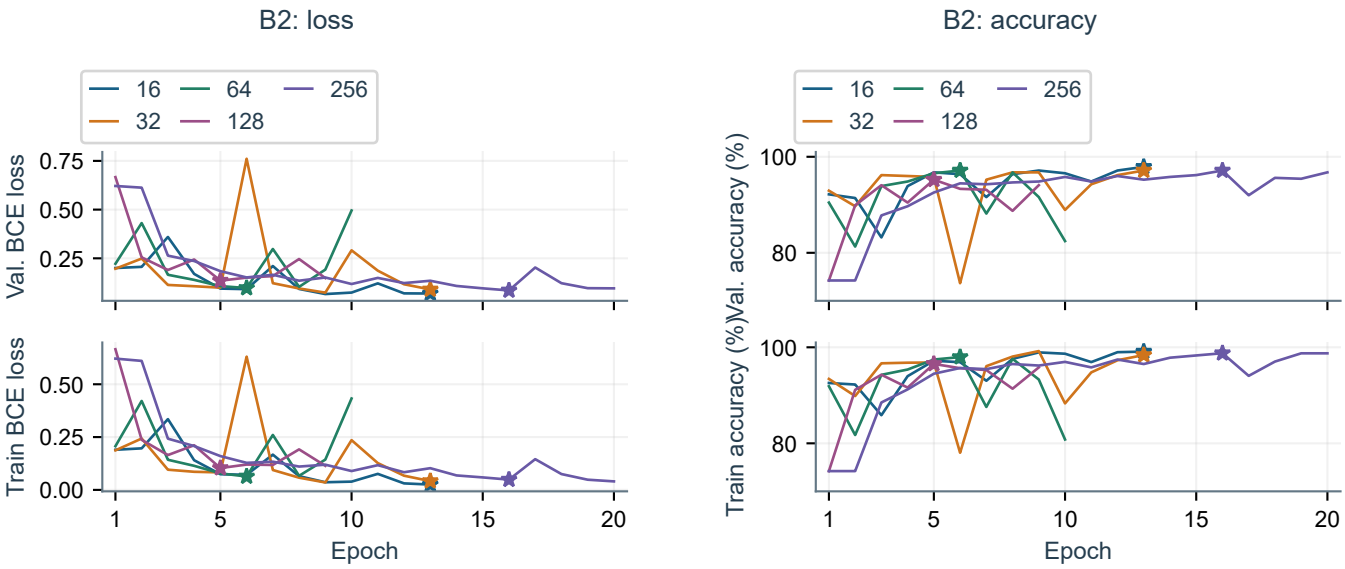


Figure B2a–b. Separate loss and accuracy figures for all five batches; validation above training. Stars mark error-selected checkpoints, not necessarily the minimum-loss epoch.

Batch 16 reaches minimum validation loss at epoch 9 (0.066596), but lowest error at 13 (0.069390); loss patience expires at 13. Its train/validation accuracies are 99.11%/97.90%. Batch 32 has a large epoch-6 loss spike/accuracy drop and then recovers, selecting 13. Batch 64 selects 6, then deteriorates through stopping at 10. Batch 128 begins near the positive-class majority rate, selects 5 and stops at 9. Batch 256 starts slowly, selects 16, and exhausts patience at the 20-epoch cap. Simultaneous train/validation reversals do not justify attributing every spike to overfitting.

Batch size also changes the update budget

For batches 16/32/64/128/256, updates per epoch are 295/148/74/37/19. Selected checkpoints receive 3,835/1,924/444/185/304 updates; batch 16 receives 12.6× as many as batch 256. Smaller batches alter gradient/BN estimates, while larger batches make fewer updates per epoch. Gradient variance and BN effects were not separately measured. This one-seed comparison selects the required training recipe, not a universal batch-size ranking or an equal-update experiment.

B2. One final test evaluation

Test accuracy	Sensitivity	Specificity	TP / TN / FP / FN
85.2564%	100.0000%	60.6838%	390 / 142 / 92 / 0

The fresh thirteen-epoch combined-data refit misses 0 positives and makes 92 false positives. Specificity is 60.6838%, versus 94.8148% for the selection checkpoint on validation. Conditional specificity is not explained by prevalence alone. The distinct refit weights, supplied train/validation overlap and absent patient IDs limit interpretation; partitions and frozen choices were preserved.

B3 · compact ResNet and architecture comparison

Use LR=0.001 and batch 16 exactly from B1/B2. Figure B3 has an initial 3×3/16 stage/max pool, then residual blocks at 16/32/64 channels; downsampling blocks use stride-2 main convolutions and 1×1 projections, with ReLU after sums. Global averaging feeds FC1. Seed 42 selects epoch 9: validation accuracy 97.5191%, loss 0.061988; loss stopping ends at 13.

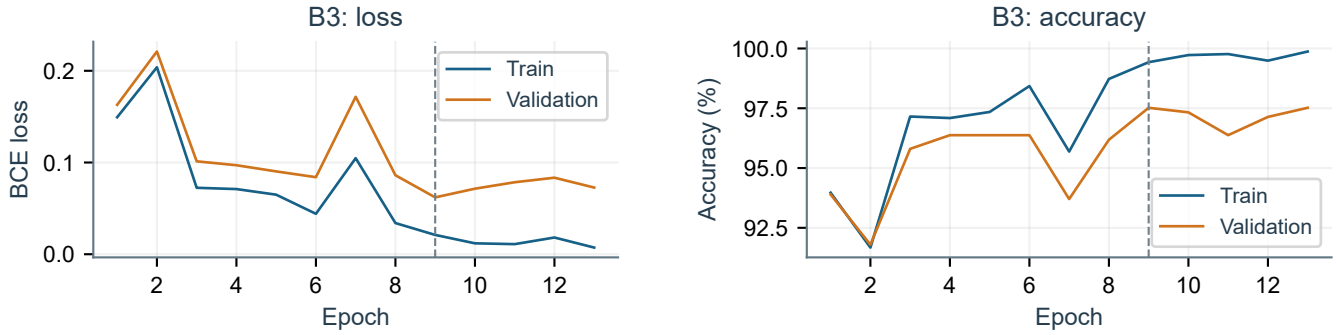


Figure B3a–b. Training/validation curves; dashed line marks selected epoch 9. The growing later gap is consistent with overfitting; earlier simultaneous fluctuations have unisolated causes.

At selection epoch 9, train/validation accuracies are 99.43%/97.52%; by epoch 13 training reaches 99.87% while validation does not improve. A fresh nine-epoch combined-data refit then detects 390/390 positives and misclassifies 118/234 negatives; specificity is 49.5726%. The table uses all three fresh refits' saved final tests without re-evaluation.

Model	Accuracy	Sensitivity	Specificity	Class. error	Parameters	FLOPs/image
B1 AlexNet	90.7051%	97.6923%	79.0598%	9.2949%	99,569	39,063,616
B2 VGG	85.2564%	100.0000%	60.6838%	14.7436%	18,185	19,464,256
B3 ResNet	81.0897%	100.0000%	49.5726%	18.9103%	77,169	25,325,696

FLOPs use one float32 1×1×64×64 evaluation input: 2 per convolution/FC multiply-add plus one per residual addition (28,672 for B3), checked against PyTorch profiler and executed-shape formulas [2]. Exclude BN, activations, pooling, FC bias addition, normalization/transfers, sigmoid/threshold and training work. These consistent scoped estimates are neither total scalar operations nor measured latency.

B1 retains a 4×4 spatial FC head and dropout, but has the most parameters/FLOPs (67,713 head parameters). B2 stacks small kernels with narrower channels/global averaging: fewest parameters/FLOPs, but compressed spatial information and no shortcuts. B3 provides residual gradient routes intended to ease deeper optimization [1], but wider channels/projections cost more than B2. Shortcut benefits were not isolated.

B1 has the highest observed accuracy and specificity. High sensitivity with lower specificity indicates many healthy images classified positive. One seed, different batches/selected epochs/update budgets, B1-only dropout and inherited tuning prevent attributing performance differences to architecture alone. The table does not trigger retuning.

[1] He et al. (2016), Deep Residual Learning for Image Recognition.

[2] PyTorch 2.11 profiler documentation (formula-based convolution/matrix FLOP estimates).

B4 · Grad-CAM on five shared random positive examples

NumPy PCG64 seed 42 uniformly samples five label-1 test rows without replacement; IDs were saved before maps, with no visual/prediction filtering. The fresh combined-data final checkpoints use main-path convolutions conv3.conv (B1), conv6.conv (B2), block64.conv2.conv (B3), all under features, before BN/ReLU; B3's shortcut is not separately mapped. Captum targets scalar pneumonia logits (target=None), evaluation mode with gradients enabled: spatially average gradients, weight channels, sum and ReLU [3,4]. Native grids 16×16/16×16/8×8 are bilinearly upsampled to 64×64 and divided by each map's own maximum.

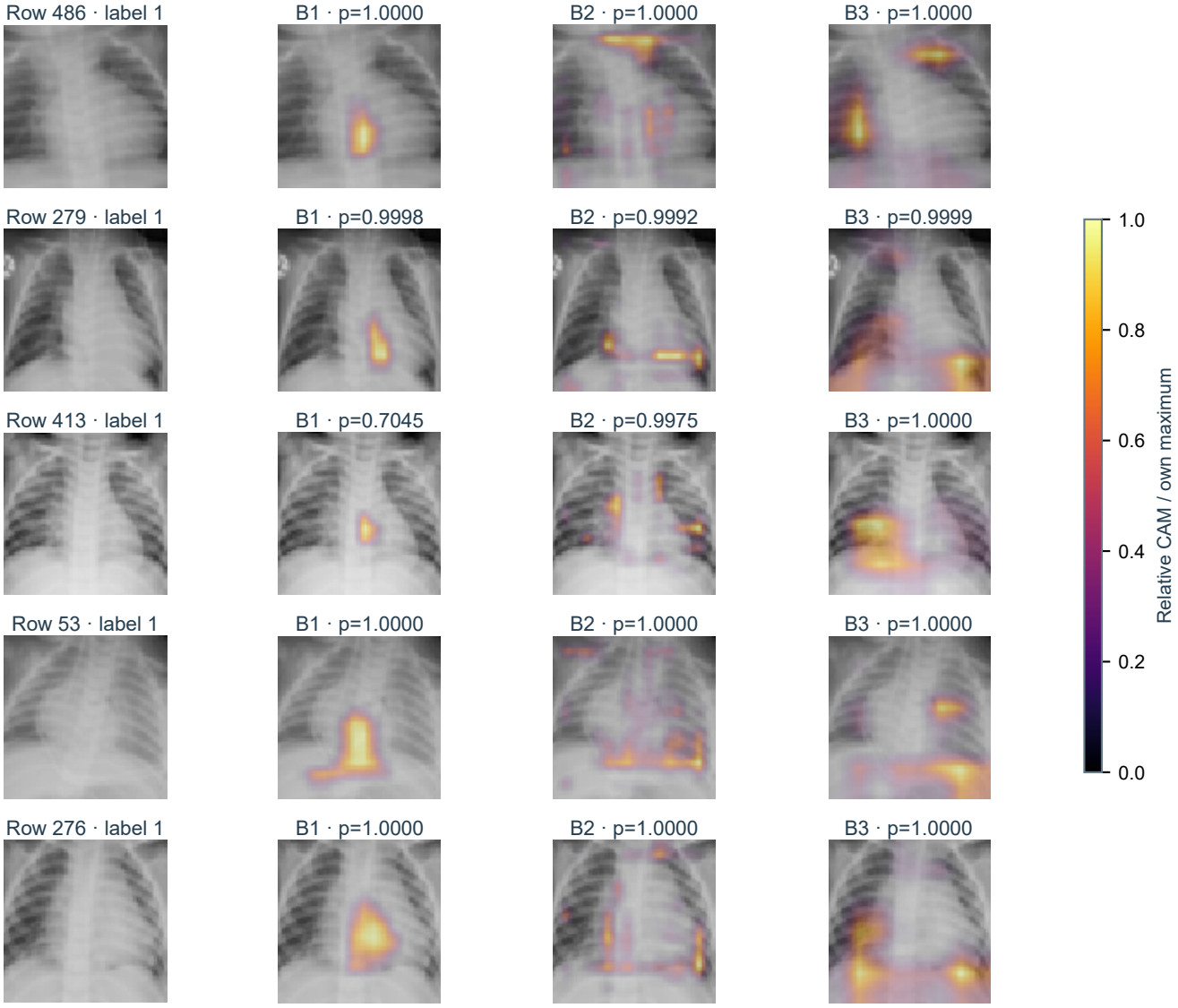


Figure B4. Original pixels at left; shared inferno and opacity=0.55×heat, no flips. p is the saved final-test probability. Each map is normalized separately; colors are relative, not absolute evidence across models.

486: B1 is lower central; B2 peaks at the upper edge; B3 highlights upper image-right and lower image-left. 279: B1 is lower central; B2/B3 favor lower boundaries. 413: B1 is central, B2 scattered, B3 broadly lower image-left. 53: B1 is lower central; B2/B3 include the lower boundary. 276: B1 overlaps central brightness; B2 is peripheral/scattered, B3 favors lower corners. Directions refer to the displayed image. Bright regions can also contain normal structures.

Mean CAM mass in the outer eight-pixel frame is 1.59%/32.35%/35.33% (B1/B2/B3), supporting this descriptive spatial comparison; it does not establish lesion localization.

Positive predictions among the five rows: 5/5/5 for B1/B2/B3. The lowest saved probability is B1 row 413: 0.7045. Coarse maps, normal structures within bright regions and absent lesion annotations prevent validating localization or causal border reliance. Five positive examples do not assess healthy/false-positive behavior. Duplicate selected-input padding matches final-test CUDA batch sizes; saved logits and independent autograd maps match exactly, with unchanged models and no full-test performance evaluation.

[3] Selvaraju et al. (2017), Grad-CAM: Visual Explanations from Deep Networks.

[4] Captum 0.9.0 LayerGradCam documentation; positive-attribution ReLU is explicit.